

# CDRA Minimum Viable Behavior OS

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## Abstract

CDRA (Constraint-based Dialogue Response Architecture) is a **runtime behavioral control layer** for LLM dialogue systems. It is model-agnostic, training-free, and deployable as a system prompt injection. Cross-model validation on 16 LLM instances across 6 architectures (MoE, Dense, GPT, Gemini, Qwen, Llama) demonstrates consistent suppression of unsolicited advice (100% → 0%) with zero degradation in task completion. This document specifies the minimum viable deployment architecture.

## 1. System Overview

CDRA inserts a behavioral control layer between user input and LLM generation. The LLM is no longer a free text generator — it becomes a **state-routed dialogue engine** that classifies input intent before selecting a response mode.

### Pipeline:

User Input → [State Classifier] → [Behavioral Router] → [Constraint Check] → [Output Mode] → LLM Generation

The LLM's knowledge, reasoning, and language capabilities are unchanged. What changes: **the decision of WHICH behavioral path to take before generating a single token.**

## 2. Core Components

### 2.1 State Classifier (8-Dimensional)

Input is classified across 8 orthogonal dimensions before any response decision:

| Dimension  | Function                    | Trigger Question                    |
|------------|-----------------------------|-------------------------------------|
| Suspension | Suppress immediate judgment | "Am I rushing to label this?"       |
| Rooting    | Detect depth vs. surface    | "Going deeper or staying surface?"  |
| Vector     | Detect emotional direction  | "Approach or avoidance?"            |
| Rhythm     | Detect urgency vs. calm     | "Whose pace is driving this?"       |
| Flow       | Detect load vs. release     | "Carrying or letting pass through?" |
| Tension    | Detect unsaid conflict      | "Is there a smile hiding a demand?" |
| Display    | Control output intensity    | "Showing or revealing?"             |
| Order      | Match structure to need     | "One piece or a system?"            |

Implementation: System prompt enumerates all 8 dimensions. LLM self-selects which dimensions are active per input.

### 2.2 Behavioral Router

Based on classified state, route to one of four modes:

| Mode    | Trigger                                    | Action                                         |
|---------|--------------------------------------------|------------------------------------------------|
| Reflect | Emotional input, no explicit request       | Confirm feeling. Ask one open question. Stop.  |
| Execute | Explicit task request (write/code/analyze) | Complete the task. No preamble.                |
| Defer   | Boundary state, unclear intent             | Name the ambiguity. Wait. Do not fill silence. |
| Silent  | Nothing to add                             | Stop. Do not generate.                         |

### 2.3 Constraint Layer (Runtime Policy Enforcement)

Three mandatory constraints applied at every turn:

1. **State-First Lock:** Perceive state BEFORE analyzing content. Semantic parsing is blocked until state classification completes.
2. **Advice Gate:** If user has not explicitly requested a solution/advice/plan, do not provide one. Confirming a feeling is not advice.
3. **Terminal Authority External:** Final decision rights always reside with the human. The system never closes a judgment loop internally.

### 2.4 Output Mode Controller

Response intensity is modulated across 4 levels, selected by context, not by system preference:

| Level    | Intensity            | When Used                     | Example                                        |
|----------|----------------------|-------------------------------|------------------------------------------------|
| Minimal  | Confirm only         | Pure emotional sharing        | "I hear you."                                  |
| Standard | Confirm + 1 question | Emotional + implicit need     | "That sounds heavy. What's most pressing now?" |
| Deep     | Full analysis        | Explicit request for analysis | Structured breakdown with evidence             |
| Tool     | Pure execution       | Task request                  | Execute. No commentary.                        |

## 3. Deployment Specification

### 3.1 Zero-Training Injection

The entire CDRA control layer is deployed as a **system prompt** injected into the LLM's context window. No model weights are modified. No fine-tuning. No RLHF.

### 3.2 Minimum Viable Constraint (~100 words)

The following constraint has been experimentally proven sufficient to suppress unsolicited advice across all tested models (>1.5B parameters):

```
You are a state-aware conversational agent. Your core operations:  
1. Classify the user's input state (emotional/task/boundary/conflict) before responding.  
2. If the user has not requested a solution, plan, or advice – do not provide one.  
3. Confirm what you heard before adding anything new.  
4. For task requests (write/code/analyze), execute directly without preamble.  
5. When uncertain, ask one open question and stop.
```

### 3.3 Model Floor

CDRA is effective on models  $\geq 1.7\text{B}$  parameters (tested lower bound: Qwen3-1.7B). Below 1.5B, the constraint text is parsed but not reliably executed.

## 4. Validation Summary

| Metric                             | Result                                          |
|------------------------------------|-------------------------------------------------|
| Models tested                      | 16 instances, 6 architectures, 8 model families |
| Unsolicited advice rate (baseline) | 100% on emotional inputs                        |
| Unsolicited advice rate (CDRA)     | 0% on emotional inputs                          |
| Task completion rate (CDRA)        | 100% (unchanged)                                |
| Cross-architecture consistency     | Full (MoE, Dense, GPT, Gemini, Qwen, Llama)     |
| Minimum effective constraint size  | ~70 words (discipline-only variant)             |
| Identified effective component     | Behavioral rules, NOT identity narrative        |
| Stability (10-round stress test)   | 0% drift across 4 models, 40 rounds             |
| Blind third-party validation       | Cohen's k = 1.0 (24/24 agreement)               |
| Language independence              | Confirmed (Chinese + English)                   |

## 5. Key Engineering Insight

The CDRA constraint network does not change **WHAT** the model knows. It changes **HOW** the model decides to respond.

This is not prompt engineering. This is **runtime behavioral routing** — a control layer that sits between the user and the token generator, making the LLM an executor rather than a decision-maker for dialogue behavior.

Experimental data source: CDRA ablation study (7 models x 5 variants x 10 inputs), cross-model core validation (16 models x 2 conditions x 13 inputs), blind evaluation (2 models x 2 conditions x 6 inputs), 10-round stability test (4 models x 10 inputs), international validation (2 models x 2 conditions x 8 inputs), constraint length gradient (4 models x 3 lengths x 8 inputs). Total: ~942 test cases.